Week 03

Version Space

Engr. Dr. Jalees Ur Rahman

Faculty Offices CS – M Block
jalees.urrahman@cust.edu.pk



Department of Computer Science
Capital University of Science and Technology, Islamabad

Topics to be covered

- Why not Find-S?
- Version Space
- Boundary Set S and G
- Candidate Elimination
- Convergence property
- Noise sensitivity
- Inductive bias
- Find-S vs Candidate Elimination
- Vocabulary

Capital University of Science & Technology
CUST
Pakistan

Recall

- Last week: Find-S returned one maximally specific consistent hypothesis.
- Problem: many hypotheses may fit the same training data.
- New idea: keep all consistent hypotheses, not just one.
- This set is called the version space.
- A hypothesis is consistent with training data if it classifies every training example correctly.

Capital University of Science & Technology
CUST
Pakistan

Version Space

- Version space = all hypotheses in H that are consistent with the data.
- $VS_{H,D} = \{h \in H \mid h \text{ is consistent with } D\}$
- It contains every rule that still remains possible.
- Version space may contain many hypotheses.
- Listing all of them can be difficult.
- So we represent it using two boundaries:
  - S: specific boundary
  - G: general boundary
- Everything valid lies between them.

ID	Income	Credit	Job	Default	Y
1	High	Good	Stable	No	1
2	Medium	Good	Stable	No	1
3	High	Poor	Stable	No	0
4	Low	Good	Unstable	No	0
5	Low	Good	Stable	No	1

- $VS_{H,D} = \{h \in H \mid \forall \langle x, c(x) \rangle \in D, h(x) = c(x)\}$
- A hypothesis stays inside the version space only if it:
  - accepts every positive example, and
  - rejects every negative example.

Boundary set S and G

- S is the set of maximally specific consistent hypotheses.
- These hypotheses are as narrow as possible while still fitting the data.
- If made any more specific, they would become inconsistent.
- G is the set of maximally general consistent hypotheses.
- These hypotheses are as broad as possible while still fitting the data.
- If made any more general, they would become inconsistent.
- Every consistent hypothesis lies between S and G.
- A valid hypothesis must be:
  - more general than some member of
  - S, and more specific than some member of G.
- So S and G together describe the whole version space.

Capital University of Science & Technology
CUST
Pakistan

Candidate Elimination

- Candidate Elimination is a supervised concept-learning algorithm.
- It updates S and G after each training example.
- Positive examples force S to become more general.
- Negative examples force G to become more specific.
- Goal: keep only hypotheses that remain consistent with all data seen so far.
- For a positive example:
  - remove inconsistent hypotheses from G
  - minimally generalize hypotheses in S so they cover the example
  - remove redundant members from S
- Positive data pushes S upward toward generalization.
- For a negative example:
  - remove inconsistent hypotheses from S
  - minimally specialize hypotheses in G so they reject the example
  - remove redundant members from G
- Negative data pushes G downward toward specificity.

Capital University of Science & Technology
CUST
Pakistan

Example

ID	Income	Credit	Job	Default	Y
1	High	Good	Stable	No	1
2	Medium	Good	Stable	No	1
3	High	Poor	Stable	No	0
4	Low	Good	Unstable	No	0
5	Low	Good	Stable	No	1

- $S_0 = \{\langle \emptyset,\emptyset,\emptyset,\emptyset \rangle\}$, $G_0 = \{\langle ?,?,?,? \rangle\}$
- $x_1 = \langle$High, Good, Stable, No$\rangle$, $c(x_1) = 1$
- $S_1 = \{\langle$High, Good, Stable, No$\rangle\}$, $G_1 = \{\langle ?,?,?,? \rangle\}$
- $x_2 = \langle$Medium, Good, Stable, No$\rangle$, $c(x_2) = 1$
- $S_2 = \{\langle ?,$ Good, Stable, No$\rangle\}$, $G_2 = \{\langle ?,?,?,? \rangle\}$
- $x_3 = \langle$High, Poor, Stable, No$\rangle$, $c(x_3) = 0$
- $S_3 = \{\langle ?,$ Good, Stable, No$\rangle\}$, $G_3 = \{\langle ?,$ Good, ?, ?$\rangle\}$
- $x_4 = \langle$Low, Good, Unstable, No$\rangle$, $c(x_4) = 0$
- $S_4 = \{\langle ?,$ Good, Stable, No$\rangle\}$, $G_4 = \{\langle ?,$ Good, Stable, ?$\rangle\}$
- $x_5 = \langle$Low, Good, Stable, No$\rangle$, $c(x_5) = 1$
- $S_4 = \{\langle ?,$ Good, Stable, No$\rangle\}$, $G_4 = \{\langle ?,$ Good, Stable, ?$\rangle\}$
- Since G_4 is consistent with this positive example, it stays. S_4 also already covers it, so it stays.

Convergence and Noise

- As more examples arrive, S and G move closer.
- If they become identical, learning has converged.
- Then the version space contains exactly one hypothesis.

- Candidate Elimination assumes noise-free data.
- Even one contradictory label can make version space empty.
- Then no hypothesis remains consistent.

- Inductive bias = assumptions a learner uses to generalize beyond seen examples.
- Candidate Elimination assumes:
  - the target concept is inside H
  - only perfectly consistent hypotheses are acceptable.
- This is also called a language bias or restriction bias.

Capital University of Science & Technology
CUST
Pakistan

Find-S vs Version Space

- Difference between Find-S algorithm and Candidate Elimination (Version Space) algorithm:

Aspect	Find-S	Candidate Elimination
Uses positive examples	Yes	Yes
Uses negative examples	No	Yes
Output	One maximally specific hypothesis	Entire version space via $S$and G
Robust to noise	No	No, very sensitive
Inductive bias	Most specific consistent rule	All consistent rules in H

Vocabulary

- **Hypothesis**: One possible classification rule.
- **Hypothesis space H**: The set of all hypotheses the learner is allowed to consider.
- **Consistent hypothesis**: A hypothesis that correctly classifies every training example seen so far.
- **Version space**: The set of all hypotheses in H consistent with the training data.
- **Specific boundary S**: The set of maximally specific consistent hypotheses.
- **General boundary G**: The set of maximally general consistent hypotheses.
- **Candidate Elimination algorithm**: The algorithm that updates S and G after each example to maintain the version space.
- **Convergence**: The point where S, G become identical, so only one consistent hypothesis remains.
- **Noise sensitivity**: The weakness that contradictory or mislabeled data can make the version space empty.
- **Inductive bias**: The assumptions that allow a learner to generalize beyond training data.

Capital University of Science & Technology
CUST
Pakistan

Quiz 1

Capital University of Science & Technology
CUST
Pakistan


Thank You!